

Quantum Teleportation

In **quantum teleportation**, the properties of **quantum entanglement** are used to send a spin state (**qubit**) between observers without physically moving the involved particle. The particles themselves are not really teleported, but the state of one particle is destroyed on one side and extracted on the other side, so the information that the state encodes is communicated. The process is not instantaneous, because information must be communicated classically between observers as part of the process. The usefulness of quantum teleportation lies in its ability to send quantum information arbitrarily far distances without exposing quantum states to thermal decoherence from the environment or other adverse effects.

Although quantum teleportation can in principle be used to actually teleport macroscopic objects (in the sense that two objects in exactly the same quantum state are identical), the number of entangled states necessary to accomplish this is well outside anything physically achievable, since maintaining such a massive number of entangled states without decohering is a difficult problem. Quantum teleportation, is, however, vital to the operation of **quantum computers**, in which manipulation of quantum information is of paramount importance. Quantum teleportation may eventually assist in the development of a "quantum internet" that would function by transporting information between local quantum computers using quantum teleportation ^[1].

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Process of Quantum Teleportation

Below is a sketch of an algorithm for teleporting quantum information. Suppose Alice has state C, which she wants to send to Bob. To achieve this, Alice and Bob should follow the sequence of steps:

1) Generate an entangled pair of electrons with spin states A and B, in a particular Bell state:

$$|\Phi_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A \otimes |\uparrow\rangle_B + |\downarrow\rangle_A \otimes |\downarrow\rangle_B).$$

Separate the entangled electrons, sending A to Alice and B to Bob.

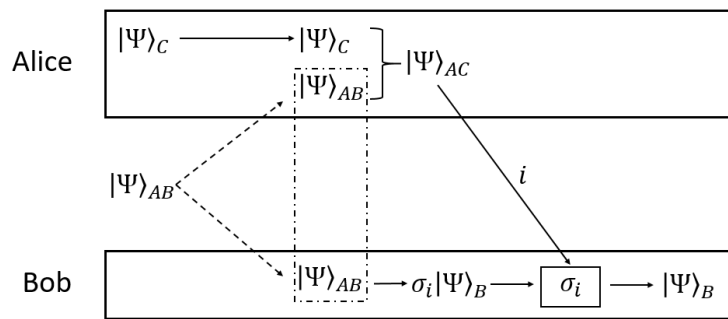
2) Alice measures the "Bell state" (described below) of A and C, entangling A and C.

3) Alice sends the result of her measurement to Bob via some classical method of communication.

4) Bob measures the spin of state B along an axis determined by Alice's measurement

Since step 3 involves communicating via some classical method, the information in the entangled state must respect causality. Relativity is not violated because the information cannot be communicated faster than the classical communication in step 3 can be performed, which is sub-lightspeed.

The idea of quantum teleportation, which can be seen in the mathematics below, is that Alice's measurement disentangles A and B and entangles A and C. Depending on what particular entangled state Alice sees, Bob will know exactly how B was disentangled, and can manipulate B to take the state that C had originally. Thus the state C was "teleported" from Alice to Bob, who now has a state that looks identical to how C originally looked. It is important to note that state C is not preserved in the processes: the **no-cloning** and **no-deletion** theorems of **quantum mechanics** prevent quantum information from being perfectly replicated or destroyed. Bob receives a state that looks like C did originally, but Alice no longer has the original state C in the end, since it is now in an entangled state with A.



Schematic of quantum teleportation algorithm. Entangled state AB is split so that particle A goes to Alice and B to Bob. Alice then entangles A and C , disentangling B and sending Bob the information he needs to recover the state of C on the other end. Based on [2].

TRY IT YOURSELF

Which of the following is true of quantum teleportation?

1) Quantum information is transferred between states

2) The teleported particle is physically transferred between locations

3) A quantum state is cloned between observers

4) Quantum information is permanently removed from the system

☐ 1 only
 ☐ 1 and 2
 ☐ 1, 2, 3, and 4
 ☐ 1 and 4
 ☐ 1 and 3
 ☐ 1, 3, and 4
 ☐ 2 only

Mathematics of Quantum Teleportation

As a review, recall the **Pauli matrices**:

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The spin operators along each axis are defined as $\frac{\hbar}{2}$ times each of $\sigma_1, \sigma_2, \sigma_3$ for the x, y, z axes respectively.

These Pauli matrices are used to construct **Bell states**, an orthonormal basis of entangled states for the tensor product space of spin- $\frac{1}{2}$ particles:

$$\begin{aligned} |\Phi_0\rangle &= I \otimes \sigma_0 |\Phi_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle) \\ |\Phi_1\rangle &= I \otimes \sigma_1 |\Phi_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle) \\ |\Phi_2\rangle &= I \otimes \sigma_2 |\Phi_0\rangle = \frac{i}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \\ |\Phi_3\rangle &= I \otimes \sigma_3 |\Phi_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\uparrow\rangle - |\downarrow\rangle \otimes |\downarrow\rangle) \end{aligned}$$

Measurements that project tensor products of spin states onto the Bell basis are called **Bell measurements**.

TRY IT YOURSELF

Which of the following correctly rewrites the state $|\uparrow\rangle \otimes |\downarrow\rangle$ in the Bell basis?

☐ $\frac{1}{\sqrt{2}}(|\Phi_1\rangle + |\Phi_2\rangle)$
☐ $\frac{1}{\sqrt{2}}(|\Phi_1\rangle - |\Phi_2\rangle)$

☐ $\frac{1}{\sqrt{2}}(|\Phi_1\rangle - i|\Phi_2\rangle)$

☐ $\frac{1}{\sqrt{2}}(|\Phi_1\rangle + i|\Phi_2\rangle)$

TRY IT YOURSELF

In the product of spin states $\frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\uparrow\rangle + |\uparrow\rangle \otimes |\downarrow\rangle)$, what is the probability that a Bell measurement will find the Bell state $|\Phi_1\rangle$?

☐ 1

☐ $\frac{1}{4}$

☐ $\frac{1}{2}$

☐ 0

Now, follow the algorithm sketched in the previous section. Suppose Alice starts with state C, which she wants to send Bob. State C can be written in the most general form:

$$|\Psi\rangle_C = c_1|\uparrow\rangle_C + c_2|\downarrow\rangle_C,$$

with c_1 and c_2 normalized complex constants.

1) Generate an entangled pair of electrons A and B in the Bell state:

$$|\Phi_0\rangle_{AB} = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A \otimes |\uparrow\rangle_B + |\downarrow\rangle_A \otimes |\downarrow\rangle_B).$$

The state of the full system of three particles is therefore $|\Psi\rangle_{ABC} = |\Phi_0\rangle_{AB} \otimes |\Psi\rangle_C$. This is a product state between entangled pair AB and non-entangled C.

2) Alice measures the Bell state of AC, entangling A and C while disentangling B. The process of measuring the Bell state projects a non-entangled state into an entangled state, since all four Bell states are entangled.

Expanding Alice's full original state, she starts with:

$$|\Psi\rangle_{ABC} = \frac{1}{\sqrt{2}}(|\uparrow\rangle_A \otimes |\uparrow\rangle_B + |\downarrow\rangle_A \otimes |\downarrow\rangle_B) \otimes (c_1|\uparrow\rangle_C + c_2|\downarrow\rangle_C)$$

Multiplying out the states and changing to the Bell basis of A and C, this state can be rewritten:

$$\begin{aligned} |\Psi\rangle_{ABC} &= \frac{1}{2}|\Phi_0\rangle_{AC} \otimes |\Psi\rangle_B + \frac{1}{2}|\Phi_1\rangle_{AC} \otimes \sigma_1|\Psi\rangle_B + \frac{1}{2}|\Phi_2\rangle_{AC} \otimes \sigma_2|\Psi\rangle_B + \frac{1}{2}|\Phi_3\rangle_{AC} \otimes \sigma_3|\Psi\rangle_B \\ &= \sum_{i=0}^3 \frac{1}{2}|\Phi_i\rangle_{AC} \otimes \sigma_i|\Psi\rangle_B \end{aligned}$$

When Alice measures the Bell state of A and C, she will find one of $|\Phi_0\rangle_{AC}$, $|\Phi_1\rangle_{AC}$, $|\Phi_2\rangle_{AC}$, $|\Phi_3\rangle_{AC}$, each with probability $\frac{1}{4}$. Whichever $|\Phi_i\rangle_{AC}$ she measures, the state of particle B will be $\sigma_i|\Psi\rangle_B$ after measurement.

3) To send Bob the state of particle C, therefore, Alice does not need to send Bob the possibly infinite amount of information contained in the coefficients c_1 and c_2 which may be real numbers out to arbitrary precision. She needs only to send the integer i of the Bell state of A and C, which is a maximum of two bits of information. Alice can send this information to Bob in whatever classical way she likes.

4) Bob receives the integer i from Alice that labels the Bell state $|\Phi_i\rangle_{AC}$ that she measured. After Alice's measurement, the overall state of the system is:

$$|\Psi\rangle_{ABC} = |\Phi_i\rangle_{AC} \otimes \sigma_i |\Psi\rangle_B.$$

Bob therefore applies σ_i to the disentangled $|\Psi\rangle_B$ state on his end, by measuring the spin along axis i . Since $\sigma_i^2 = I$ for all i , Bob is left with the overall state:

$$|\Psi\rangle_{ABC} = |\Phi_i\rangle_{AC} \otimes |\Psi\rangle_B$$

Bob has therefore changed the spin state of particle B to:

$$|\Psi\rangle_B = c_1 |\uparrow\rangle_B + c_2 |\downarrow\rangle_B,$$

which is identical to the original state of particle C that Alice wanted to send. The information in state C has been "teleported" to Bob's state: the final spin state of B looks like C's original state. Note, however, that the particles involved never change between observers: Alice always has A and C, and Bob always has B.

References

1. Pirandola, S., & Braunstein, S. *Physics: Unite to build a quantum Internet*. Retrieved from <http://www.nature.com/news/physics-unite-to-build-a-quantum-internet-1.19716>

2. Debenben, . *quantum teleportation diagram*. Retrieved from <https://commons.wikimedia.org/w/index.php?curid=34503176>

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